Swetha Vaidyanathan Sriram

Software Engineer | Full-Stack & AI/ML

vsswetha@gmail.com | 925-750-0169 | PA (Open to Relocation) | swetha-sriram | swetha7502 | Portfolio

EDUCATION

Master of Science, Computer Science — **2025 – 2027** | **CGPA: 4/4**
The Pennsylvania State University, Pennsylvania

Bachelor of Engineering, Computer Science and Engineering — **2020 – 2024** | **CGPA: 3.94/4**
Madras Institute of Technology, Anna University, India

Bachelor of Science, Data Science & Programming (Hybrid) — **2020 – 2024** | **CGPA: 3.73/4**
Indian Institute of Technology, Madras, India

SKILLS

Languages — Python, Java, C++/C, JavaScript/TypeScript, C#, Bash, SQL

Databases — PostgreSQL, MySQL, MongoDB, Redis, SQLite

Web & Backend — FastAPI, Flask, ReactJS, Vue.js, Next.js, Node.js, HTML5, CSS3, REST APIs

AI & ML — LangSmith, LangChain, LangGraph, PyTorch, ChromaDB, OpenCV, TensorFlow, GitHub Copilot, Claude Code

Tools & Platforms — AWS, GCP, Linux, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Celery, CI/CD, Oracle APEX

Core CS — Data Structures & Algorithms, System Design, OOP, Git

WORK EXPERIENCE

AI Solutions Intern | Penn State - Global Engagement Office — **Jul 2026 – Present**
- Built a **full-stack** event management platform (**React/TypeScript, Express/Node, Supabase/PostgreSQL**) with an AI planning assistant (**OpenAI** function-calling) grounding predictions in historical data, and a live-syncing Kanban board with role-based approval workflows.
- Solo-owned schema/API design, security, and prod deployment using **Claude Code** for accelerated development, shipping a live system used by campus coordinators.

Research Assistant | Pennsylvania State University — **Jan 2026 – May 2026**
- Accelerated molecular dynamics simulations by 15%, analyzing 10,000+ trajectories across 50 protein complexes using **C#** and **MATLAB**.

Associate Applications Developer | Oracle Pvt. Ltd. — **Aug 2024 – Aug 2025**
- Enhanced an enterprise-scale release management platform (100+ users across APAC and EMEA) by **designing and deploying 2 new workflows** using **PL/SQL, JavaScript, and Oracle APEX**, reducing manual scheduling (~20% improvement).
- Designed and developed a **Jira ticket dashboard** (150+ DAUs) with real-time status visibility, reducing weekly status meetings by ~20% and accelerating ticket resolution time by 5%
- Deployed **3 Oracle SDK releases**, implementing core features, bug fixes and delivery of enterprise updates.
- Resolved critical bugs and enhancements in **Java** and developed **JUnit** tests, improving system reliability
- Analyzed and optimized **Oracle Cloud Infrastructure** (OCI) logs to identify and eliminate redundant entries.

Project Intern | Oracle Pvt. Ltd. — **Jan 2024 – Jul 2024**
- **Automated** regression **testing flows** with Oracle Utilities Testing Accelerator (**OUTA**) and **Selenium**, boosting test coverage by ~30% and reducing manual QA effort by ~40%
- Built and deployed an energy forecasting model adopted by 10+ data science team members using **Oracle ADB, Python, Java and Selenium**

ACADEMIC PROJECTS

Rollback-Safety Prover | Python, FastAPI, PostgreSQL, Alembic, Docker, GitLab CI/CD
- Built a CI proof harness that runs prior app versions from git worktrees against new-version data on disposable PostgreSQL, catching unsafe Alembic migrations via a 4-test compatibility matrix.
- Automated GitLab CI/CD with Docker builds, SAST, secret detection, and a merge-blocking proof gate; deploys auto-roll back to the last proven revision on failed health checks.

Escrow | Python, React, WebSockets, Docker, NVIDIA NemoClaw | Nebius x NVIDIA Hackathon 2026
- Built transactional middleware (Python, NVIDIA NemoClaw) that intercepts AI agent tool calls, stages side effects in an outbox, and commits only after policy checks pass.
- Restricted sandbox egress to a single endpoint with host-side sends; streamed live event flow and action approvals to a React + WebSockets dashboard.

CodeBase Intel | Python, FastAPI, PostgreSQL, Celery, Redis, React, Terraform, GCP
- Built a polyglot tree-sitter AST parser that models cross-file dependencies in PostgreSQL (Supabase) with recursive CTEs and renders an interactive graph via React Flow + Vite; async ingestion offloaded to Celery + Redis
- Enabled semantic code search and RAG-based Q&A using ChromaDB + OpenAI; deployed to GCP Cloud Run with multi-stage Docker builds, Terraform IaC, and GitHub Actions CI/CD